British Math Olympiad Round 1 2004 — P5/5

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Let p, q and r be prime numbers. It is given that p divides $qr - 1$, q divides $pr - 1$, and r divides $pq - 1$. Determine all possible values of pqr.

Solution

Firstly if $p = q = r$ then $p \mid p^2 - 1 = (p-1)(p+1)$ but this is impossible since $\gcd(p, p+1) = \gcd(p, p-1) = 1$. Also due to symmetry in (p, q, r) we can say, without loss of generality, if $p = q$ then $p \mid pr - 1$ but that means that $p \mid (-1)$ and this is impossible since prime numbers are larger than 1.

Thus it is possible to assume, without loss of generality, $p < q < r$.

Claim 1: $p = 2$.

Proof: Assume not, then $3 \leq p < q < r$. Since $pqr \mid (pq-1)(rp-1)(qr-1) = (pqr)^2 - pqr(p+q+r) + (pq+qr+rp) - 1$, we must have $pq + qr + pr \equiv 1 \pmod{pqr}$. Now by assumption, $pq + qr + rp < 3qr \leq pqr$. That means that $pq + qr + rp = 1$, since it can't be more than pqr. However since all of the primes are ≥ 3, this is a contradiction. □

Claim 2: $q = 3$.

Proof: Again assume not, then $q \geq 4$. Due to claim 1, we must have $q \mid (2r-1)$ and $r \mid (2q-1)$, thus $qr \mid (2r-1)(2q-1) = 4qr - 2(q+r) + 1$, but for this to be true we need $2(r+q) \equiv 1 \pmod{qr}$. Now by assumption, $2(r+q) < 4r \leq qr$ but this means that $2(p+q) = 1$ since it can't be bigger than qr. But this is impossible since p and q are positive integers. □

Now we finish the problem by using claims 1 and 2. We must have a prime r dividing $2 \cdot 3 - 1 = 5$, and that forces $r = 5$. Thus the only possible value of pqr is $2 \cdot 3 \cdot 5 = 30$. ■